

Secretary Problem Research

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Lecture 1: abstract error SP

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1 Abstract Error SP

1.1 Basic Definitions

$i \approx j$ means that i and j are in-distinguishable in the first observation.

- (reflexivity) $i \approx i$
- (symmetry) $i \approx j \Leftrightarrow j \approx i$
- (connectedness) $i \approx j, r_i < r_j \Rightarrow \forall k, l [r_k, r_l \in [r_i, r_j] \Rightarrow r_k \approx r_l]$

1.2 Algorithmic Representations

Algorithm 1 Oct 14 2021

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1: for  $i \in [n]$  do
2:    $\alpha \leftarrow$  the current item to be decided
3:   if  $i < r$  then  $\triangleright$  Phase 1
4:     if  $\exists \beta \in O_i. \alpha <_i \beta$  then  $\triangleright \alpha$  is not a maximum
5:       Decline this item
6:     else if  $\alpha \in E_i$  then
7:       Decline this item
8:     else
9:       Examine this item
10:    end if
11:  else if  $i < n$  then  $\triangleright$  Phase 2
12:    if  $\exists \beta \in O_i. \alpha <_i \beta$  then
13:      Decline this item
14:    else
15:      if  $\forall \beta \in O_i. \beta \leq_i \alpha$  then  $\triangleright \alpha$  is the absolute maximum
16:        Accept this item
17:      else if  $\alpha \in E_i$  then
18:        Decline this item
19:      else
20:        Examine this item
21:      end if
22:    end if
23:  else
24:    Accept this item
25:  end if
26: end for
```

Algorithm 2 Game Process, Relaxed

```

1:  $[n], r_i$  and  $\approx$  relation are given according to definition
2:  $O_1 = \{1\}$ 
3:  $E_1 = \emptyset$ 
4:  $\alpha_1 = 1$ 
5:  $\Theta_1 = \emptyset$ 
6: for  $i \in [n]$  do
7:    $d_i \leftarrow$  player's decision
8:   if  $d_i = \text{Accept}$  then
9:     Game terminates with reward  $v(\alpha_i)$ 
10:  else if  $d_i = \text{Decline}$  then
11:     $O_{i+1} \leftarrow O_i \cup \{i+1\}$ 
12:     $E_{i+1} \leftarrow E_i$ 
13:     $\alpha_{i+1} \leftarrow i+1$ 
14:  else if  $d_i = \text{Examine}$  then
15:     $O_{i+1} \leftarrow O_i$ 
16:     $E_{i+1} \leftarrow E_i \cup \{i\}$ 
17:     $\alpha_{i+1} \leftarrow \alpha$ 
18:  end if
19:   $\Theta_{i+1} \leftarrow \{(\beta, \gamma) \in O_{i+1}^2 : r_\beta \leq r_\gamma, \beta \not\approx \gamma\} \cup \{(\beta, \gamma) \in E_{i+1}^2 : r_\beta \leq r_\gamma\}$ 
20: end for

```

1.3 Calculation under a relaxed condition

We give the expected value of ALG on the relaxed model (only one examine is required to obtain precise comparisons). Let α denote the selected item.

$$\mathbb{E}[\text{OPT}] = m \quad (1)$$

$$\mathbb{E}[\text{ALG}] = \sum_{t=1}^m t \Pr[v(\alpha) = t] \quad (2)$$

$$\Pr[v(\alpha) = t] = \sum_{i=r}^n \Pr[\alpha = i \mid v(i) = t] \Pr[v(i) = t] \quad (3)$$

$$+ \Pr[\text{reach } n \cap \text{not accept } n] \Pr[v(n) = t] \quad (4)$$

$$+ \Pr[\text{reach } n-1 \cap \text{ED } n-1] \Pr[v(n-1) = t] \quad (5)$$

The two added terms are cases where the items are exhausted. If n is reached then it must be selected. If $n-1$ is examined then it must be selected also.

$$\Pr[v(i) = t] = 1/m, \forall i \forall t \quad (6)$$

$$\Pr[\alpha = i \mid v(i) = t] = \Pr[\text{reach } i \mid \max_{j < r} v(j) \leq t] \Pr[\max_{j < r} v(j) \leq t] \quad (7)$$

$$= \sum_{s=1}^t \Pr[\text{reach } i \mid \max_{j < r} v(j) = s] \Pr[\max_{j < r} v(j) = s] \quad (8)$$

$$\Pr[\text{reach } n \cap \text{not accept } n] = \sum_{s=t+1}^m \Pr[\text{reach } n \mid \max_{j < r} v(j) = s] \Pr[\max_{j < r} v(j) = s] \quad (9)$$

$$\Pr[\text{reach } n-1 \cap \text{ED } n-1] = \sum_{s=t+1}^m \mathbf{1}_{s \leq t+\delta} \Pr[\text{reach } n-1 \mid \max_{j < r} v(j) = s] \Pr[\max_{j < r} v(j) = s] \quad (10)$$

$$\Pr[\max_{j < r} v(j) = s] = \Pr[\max_{j < r} v(j) \leq s] - \Pr[\max_{j < r} v(j) \leq s-1] \quad (11)$$

$$= \left(\frac{s}{m}\right)^{r-1} - \left(\frac{s-1}{m}\right)^{r-1} \quad (12)$$

$$\Pr[\text{reach } i \mid \max_{j < r} v(j) = s] \quad (13)$$

$$= \sum_{(a_1 \dots a_k) \in \{1,2\}^k, \sum a_j = i-r} \prod_{j=1}^k \mathbf{1}_{\delta < t} \left(\mathbf{1}_{a_j=1} \frac{s-\delta-1}{m} + \mathbf{1}_{a_j=2} \frac{\delta}{m} \right) + \mathbf{1}_{\delta \geq t} \frac{t-1}{m} \quad (14)$$

$$= \sum_{x=0}^{\lfloor \frac{i-r}{2} \rfloor} \left(\mathbf{1}_{\delta < t} \left(\frac{\delta}{m} \right)^x \left(\frac{s-\delta-1}{m} \right)^{i-r-2x} + \mathbf{1}_{\delta \geq t} \left(\frac{t-1}{m} \right)^x 0^{i-r-2x} \right) \times \frac{(i-r-x)!}{x!(i-r-2x)!} \quad (15)$$

Explanation: There are $i - r$ steps to go from r to i . Each decline will go 1 step, and each examine+decline will go 2 steps. We calculate the probability to take each possible path and then sum them up. In the second equation, x is the number of items examined going from r to i .

Lecture 2: Relaxed SP

Sun 30 Jan 2022 15:50

1.4 relaxed model PI

1.4.1 calculation 1 (in meeting)

$$\begin{aligned}
& \sum_{i=1}^n \Pr[\text{accept } i \cap i \text{ best}] \\
&= \sum_{i=1}^n \Pr[\text{accept } i \mid i \text{ best}] \Pr[i \text{ best}] \\
&= \frac{1}{n} \sum_{i=1}^n \Pr[X_1, \dots, X_{i-1} < T \cap X_i \geq T \mid i \text{ best}] \\
&= \frac{1}{n} \sum_{i=1}^n \Pr[X_1, \dots, X_{i-1} < T \mid X_i \geq T \cap i \text{ best}] \Pr[X_i \geq T \mid i \text{ is best}]
\end{aligned}$$

Note that $\Pr[X_i \geq T \mid i \text{ best}] = 1 - F_{X_{\max}}(T) = 1 - T^n$. Also,

$$\begin{aligned}
& \Pr[X_1, \dots, X_{i-1} < T \mid X_i \geq T \cap i \text{ best}] \\
&= \int_T^1 \Pr[X_1, \dots, X_{i-1} < T \mid i \text{ best} \cap X_i = x] f_X(x \mid X_i \geq T \cap i \text{ best}) dx \\
&= \int_T^1 \Pr[X_1, \dots, X_{i-1} < T \mid X_1, \dots, X_{i-1} < x] f_{X_{\max}}(x \mid X_{\max} \geq T) dx \\
&= \int_T^1 \left(\frac{T}{x}\right)^{i-1} \frac{nx^{n-1}}{1-T^n} dx \\
&= \frac{nT^{i-1}}{1-T^n} \int_T^1 x^{n-i} dx \\
&= \frac{nT^{i-1}}{1-T^n} \frac{1-T^{n-i+1}}{n-i+1}
\end{aligned}$$

Combining all terms,

$$\begin{aligned}
& \sum_{i=1}^n \Pr[\text{accept } i \cap i \text{ best}] \\
&= \frac{1}{n} \sum_{i=1}^n \frac{nT^{i-1}}{1-T^n} \frac{x^{n-i+1}}{n-i+1} (1-T^n) \\
&= \sum_{i=1}^n T^{i-1} \frac{1-T^{n-i+1}}{n-i+1}
\end{aligned}$$

1.4.2 Calculation 2

We have n items with weights $X_1, \dots, X_n \sim \text{Unif}(0, 1)$. Observed values are $\tilde{X}_i \sim \text{Unif}(X_i - \delta, X_i + \delta)$. We adopt a thresholding scheme for ALG: We examine an item if $\tilde{x}_i > T$ and accept if $x_i > T$.

If the player reaches X_n , then X_n must be accepted.

$$\begin{aligned}
\Pr[X_{\max} \text{ selected}] &= \sum_{i=1}^{n-1} \Pr[X_1, \dots, X_{i-1} < T \cap X_i \geq T \cap X_i \text{ is max}] \\
&\quad + \Pr[X_1, \dots, X_{n-1} < T \cap X_n \text{ max}] \\
&= \sum_{i=1}^{n-1} \int_0^1 \Pr[X_1, \dots, X_{i-1} < T \cap X_i \geq T \cap X_{i+1}, \dots, X_n \leq X_i \mid X_i = x] f_{\text{Unif}(0,1)}(x) dx \\
&\quad + \Pr[X_n \text{ max}] \int_0^1 \Pr[X_1, \dots, X_{n-1} < T \mid X_n = x \text{ is max}] f_{\text{max Unif}(0,1)}(x) dx \\
&= \sum_{i=1}^{n-1} \int_T^1 \Pr[X_1, \dots, X_{i-1} < T \cap X_{i+1}, \dots, X_n \leq x] f_{\text{Unif}(0,1)}(x) dx \\
&\quad + \frac{1}{n} \int_0^1 \Pr[X_i < T \mid X_i < x]^{n-1} f_{\text{max Unif}(0,1)}(x) dx \\
&= \sum_{i=1}^{n-1} \int_T^1 T^{i-1} x^{n-i} dx + \int_0^T x^{n-1} dx + \int_T^1 x^{n-1} \left(\frac{T}{x}\right)^{n-1} dx \\
&= \sum_{i=1}^{n-1} \frac{1}{n-i+1} T^{i-1} (1 - T^{n-i+1}) + \frac{1}{n} T^n + T^{n-1} (1 - T)
\end{aligned}$$

2 Relaxed SP Prophet Inequality

2.1 Lower Bound of $P^{\max}$

In this section, we provide a lower bound of $P^{\max}$, where

$$P^{\max} = \max_{T \in [0,1]} \sum_{i=1}^n \frac{1}{n-i+1} T^{i-1} (1 - T^{n-i+1}).$$

(Since we are interested in asymptotic behavior as $n \rightarrow \infty$, the difference in the last term can be safely ignored.)

The optimal T^* can be solved by the optimization problem. Let's show the

lower bound of $P^{\max}(T^*)$. Let $T = \frac{n}{n+1}$, then we have

$$\begin{aligned} P^{\max}(T^*) &\geq P^{\max}(T) \\ &= \sum_{i=1}^n \frac{1}{n-i+1} T^{i-1} (1 - T^{n-i+1}) \\ &= \sum_{i=1}^n \frac{1}{n-i+1} T^{i-1} - \sum_{i=1}^n \frac{1}{n-i+1} T^n \\ &= T^n \sum_{i=1}^n \frac{1}{i} (T^{-i} - 1) \end{aligned}$$

We take the limit $n \rightarrow \infty$, we have

$$\lim_{n \rightarrow \infty} T^n \sum_{i=1}^n \frac{1}{i} (T^{-i} - 1) = \frac{1}{e} \lim_{n \rightarrow \infty} \sum_{i=1}^n \frac{1}{i} \left(\left(1 + \frac{1}{n}\right)^i - 1 \right)$$

Observe that

$$\left(1 + \frac{1}{n}\right)^i - 1 = \sum_{k=1}^i \binom{i}{k} \frac{1}{n^k}.$$

Then, we take the sum on i , we have

$$\sum_{i=1}^n \frac{1}{i} \left(1 + \frac{1}{n}\right)^i - 1 = \sum_{i=1}^n \frac{1}{i} \sum_{k=1}^i \binom{i}{k} \frac{1}{n^k}$$

By Fubini Theorem, we have

$$\sum_{i=1}^n \frac{1}{i} \sum_{k=1}^i \binom{i}{k} \frac{1}{n^k} = \sum_{k=1}^n \sum_{i=k}^n \frac{1}{i} \binom{i}{k} \frac{1}{n^k}$$

We have

$$\frac{1}{i} \binom{i}{k} = \frac{(i-1)(i-2)\dots(i-k+1)}{k!}$$

The nominator is a polynomial with order $k-1$ with respect to i .

Exercise 2.1. Show that $\sum_{i=1}^n i^k = \frac{1}{k+1} n^{k+1} + O(n^k)$

By this property, we have

$$\lim_{n \rightarrow \infty} \sum_{i=k}^n \frac{1}{i} \binom{i}{k} \frac{1}{n^k} = \frac{1}{k \cdot k!} \frac{n^k}{n^k} = \frac{1}{k \cdot k!}.$$

Finally, we have the lower bound of $P^{\max}$ is

$$\frac{1}{e} \sum_{k=1}^n \frac{1}{k \cdot k!} \approx 0.485.$$

2.2 Upper Bound of $P^{\max}$

Consider there are only two secretaries in the arriving sequence. We have $X_1 \sim \text{Unif}(0, 1)$, $X_2 \sim \text{Unif}(0, 1)$.

Exercise 2.2. Show that the upper bound of $P^{\max}$ among all deterministic algorithms is $\frac{1}{2}$

Hint: Any deterministic algorithm can only choose X_1 or X_2 . If $X_1 = x$, what will happen? Then take integral. Another method is that think about a coupling random arrival instance $Y_1 = X_1$, $Y_2 = \text{Unif}(0, 1)$. Y obviously follows the arrival process distribution. When the first customer arrives, any algorithm cannot distinguish the arrival instance is X or Y . Then for the first customer, if the algorithm accepts him, ..., if not accept,....

Exercise 2.3. Show that the upper bound of $P^{\max}$ among all randomized algorithms is $\frac{1}{2}$

Hint: Yao's principle. (or set a probability distribution on X_1 and X_2).

Lecture 3: response to excercises

Mon 07 Feb 2022 00:24

2.3 Responses

2.3.1 Response to (2.1)

Proof. Use induction on n . The base case is trivial:

$$1 = \frac{1}{k+1} + O(1).$$

Now suppose the property holds for n .

$$\begin{aligned} \sum_{i=1}^{n+1} i^k &= (n+1)^k + \sum_{i=1}^n i^k = (n+1)^k + \frac{1}{k+1} n^{k+1} + O(n^k) \\ \frac{1}{k+1} (n+1)^{k+1} &= \frac{1}{k+1} n^{k+1} + O(n^k). \end{aligned}$$

By considering $(n+1)^k = O(n^k)$ and $O(n^k) \pm O(n^k) = O(n^k)$, the difference of the right hand side of the above equations is $O(n^k)$. The left hand side shows the property for $n+1$. $\square$

2.3.2 Response to (2.2)

Optimal P for a determininstic algorithm is 0.75.

Proof. ALG is competely characterized by a function $\alpha : [0, 1] \rightarrow [0, 1]$, where $\alpha(x)$ represents the probability of accepting X_1 after an observation $X_1 = x$.

$$\begin{aligned}
 \Pr[\text{win}] &= \int_0^1 \alpha \Pr[X_2 < x] + (1 - \alpha) \Pr[X_2 > x] f(x) dx \\
 &= \int_0^1 \alpha x + (1 - \alpha)(1 - x) dx \\
 &= \int_0^1 2\alpha x - \alpha dx + \frac{1}{2} \\
 &\leq \max_{\alpha} \int_0^1 2\alpha x - \alpha dx + \frac{1}{2}.
 \end{aligned}$$

ALG is deterministic iff $\alpha(x) \in \{0, 1\}$. If α takes 0, the integral is zero. If α takes 1, $2\alpha x - \alpha > 0$ iff $x > \frac{1}{2}$. Hence to maximize the integral we take $\alpha(x) = \begin{cases} 1 & x \geq \frac{1}{2} \\ 0 & x < \frac{1}{2} \end{cases}$, which gives an upper bound of $\frac{3}{4}$. $\square$

Response to (2.3)

Lecture 4: Relaxed SP with error

Tue 01 Feb 2022 00:06

3 Calculation with error involved

The weights $X_1, \dots, X_n \sim \text{Unif}(0, 1)$. The noisy observations $\tilde{X}_1, \dots, \tilde{X}_n \sim \text{Unif}(x_i - \delta, x_i + \delta)$.

The strategy is to introduce thresholding parameters R and $T \in (0, 1)$. We examine the upcoming item i if $\tilde{x}_i > R$, and accept the item if $x_i > T$. We still want to find $\Pr[X_{\max}$ is selected] given particular R, T, n, δ .

$$\begin{aligned}
 \Pr[X_{\max} \text{ selected}] &= \sum_{i=1}^{n-2} \int_0^1 \Pr[X_i \text{ best} \cap i \text{ selected} \mid X_i = x] f(x) dx \\
 &\quad + \Pr[\text{reach } n-1 \cap X_{n-1} \text{ best} \cap \tilde{X}_{n-1} > R] \\
 &\quad + \Pr[\text{reach } n \cap X_n \text{ best}]
 \end{aligned} \tag{16}$$

For all $i \leq n-2$:

$$\begin{aligned}
 &\Pr[X_i \text{ best} \cap i \text{ selected} \mid X_i = x] \\
 &= \Pr[\text{reach } i \cap \tilde{X}_i > R \cap X_i > T \cap X_i \text{ max} \mid X_i = x] \\
 &= \Pr[\text{Unif}(x - \delta, x + \delta) > R] \mathbf{1}(x > T) \Pr[\text{reach } i \cap X_i \text{ max} \mid X_i = x]
 \end{aligned}$$

For all $i \leq i - 2$:

$$\begin{aligned}
 & \Pr[\text{reach } i \cap X_i \text{ max} \mid X_i = x] \\
 &= \sum_{k=0}^{\lfloor \frac{i-1}{2} \rfloor} \Pr[\text{reach } i \cap X_i \text{ max} \cap k \text{ examines before reaching } i \mid X_i = x] \\
 &= \sum_{k=0}^{\lfloor \frac{i-1}{2} \rfloor} \Pr[\text{reach } i \text{ with } k \text{ examines}] \Pr[X_i \text{ max} \mid \text{reach } i \text{ with } k \text{ examines} \cap X_i = x] \\
 &= \sum_{k=0}^{\lfloor \frac{i-1}{2} \rfloor} \underbrace{\frac{(i-1-k)!}{k! (i-1-2k)!}}_{\text{ways to reach } i} \Pr[\text{ED}]^k \Pr[\text{D}]^{i-1-2k} \underbrace{F(x|X < T)^k}_{\text{examined}} \underbrace{F(x)^k}_{\text{omitted}} \underbrace{F(x|\tilde{X} < R)^{i-1-2k}}_{\text{declined directly}} \underbrace{F(x)^{n-i}}_{\text{upcoming}}
 \end{aligned}$$

where

$$\begin{aligned}
 \Pr[\text{ED}] &= \frac{1}{2\delta} \int_0^T \int_{y-\delta}^{y+\delta} \mathbf{1}(z > R) dz dy = \text{const} \\
 \Pr[\text{D}] &= \frac{1}{2\delta} \int_0^1 \int_{y-\delta}^{y+\delta} \mathbf{1}(z < R) dz dy = \text{const}
 \end{aligned}$$

and for $0 \leq x \leq 1$:

$$\begin{aligned}
 F(x|X < T) &= \begin{cases} \frac{x}{T} & 0 \leq x < T \\ 1 & x \geq T \end{cases} \\
 F(x|\tilde{X} < R) &= \frac{\int_0^x \int_{y-\delta}^{y+\delta} \mathbf{1}(z < R) dz dy}{\int_0^1 \int_{y-\delta}^{y+\delta} \mathbf{1}(z < R) dz dy} \\
 F(x) &= x.
 \end{aligned}$$

4 Another Approach to Estimate $P^{\max}$

To simplify the writing, we write Error Win as the event that we find the best secretary under the model with error. We write Original Win as the event that we find the best secretary under the original model.

$$\begin{aligned}
 \Pr(\text{Error Win}) &= \Pr(\text{Error Win} | \text{Original Win}) \Pr(\text{Original Win}) \\
 &\quad + \Pr(\text{Error Win} | \text{Original Lose}) \Pr(\text{Original Lose}) \\
 &\geq 0.485 \Pr(\text{Error Win} | \text{Original Win}).
 \end{aligned}$$

Optional exercise: Is there any method to bound $\Pr(\text{Error Win} | \text{Original Lose})$.

Next, we bound the probability of Error win conditional on Original Win. We start with defining two events: we let $\mathcal{E}_x = \max X_i = x$. We let $\mathcal{F}_y$ be that the secretary who arrives in the last period before the max secretary arrives. Because the arriving process is i.i.d, it is easy to see that $\mathcal{F}_y \sim \text{Unif}(0, 1)$.

Exercise1: Find the distribution of $\mathcal{E}_x$.

Hint: See Beta distribution.

We have

$$\Pr(\text{Error Win}|\text{Original Win}) = \int_{x,y} \Pr(\text{Error Win}|\text{Original Win}, \mathcal{E}_x, \mathcal{F}_y) f(x) dx dy.$$

We construct two arriving instances as follows: let I_1 contains n i.i.d $\text{Unif}(0, 1)$ r.v. In additional, the maximal realization is x . The last r.r. before the max one has realization y . Let I_2 be a copy of I_1 . Conditional on Original win, there are two cases that we do not win on the same instance:

- The max x has observation $\hat{x} < T$, then we do not examine the best secretary.
- The one before max, y , has observation $\hat{y} \geq T$, then the best secretary is skipped. (Notice that here we do not need to consider the second last before x and all the previous r.v., because we are providing the lower bound.)

Therefore, we have

$$\Pr(\text{Error lose}|\text{Original Win}, \mathcal{E}_x, \mathcal{F}_y) \leq \Pr(\text{Case 1} \cup \text{Case 2}) \leq \Pr(\text{Case 1}) + \Pr(\text{Case 2}).$$

Exercise 2: Find the probability of Case 1 and Case 2 by considering that the observation cannot exceed 1 or below 0.

All calculation were based on the assumption that $\delta < T < 1 - \delta$.

$$F(x) = \begin{cases} 1 & x < T - \delta \\ 0 & x > T + \delta \\ \frac{x + \delta - T}{2\delta} & \text{otherwise} \end{cases}$$

$$\Pr(\text{Case 1}) = F(x)$$

$$\Pr(\text{Case 2}) = 1 - F(y)$$

Finally, we have

$$\begin{aligned} \Pr(\text{Error Win}) &\geq 0.485 \Pr(\text{Error Win}|\text{Original Win}) \\ &\geq 0.485 \int_{x,y} (1 - \Pr(\text{Case 1}) - \Pr(\text{Case 2})) f(x) dx dy. \end{aligned}$$

Please complete the final estimation. Write it as a function of δ , if the integral is hard to calculate, run numerical experiments for different δ and see what the value of the integral is.

Evaluated Integral:

$$\frac{((-2n-1)\delta - T)(T - \delta)^n + ((-2n-1)\delta + T)(\delta + T)^n + 2T\delta(n+1)}{2(n+1)\delta} \xrightarrow{n \rightarrow \infty} T$$

$$\begin{aligned} \Pr[\text{Error Win} \mid \text{Original Lose}] &= \Pr[\text{reach } i \mid \exists j < i, X_j > T] \Pr[X_{\max} > T \mid \text{Original Lose}] \\ &\leq \Pr[j-1 \text{ examine}] \Pr[X_{\max} > T \mid \text{Original Lose}] \end{aligned}$$

Lecture 5: Properties of the optimal solution to the relaxed problem with and without error

Wed 09 Mar 2022 23:23

5 Simple Problem

We begin with a relaxed problem of best selection with no error. Let $[x, n]$ denote the initial game state where we have an upcoming item with value x and n un-observed items after the current item.

If the player accepts the first item, he wins iff all other items are less than the first item. Hence $\Pr[\text{win} \mid \text{accept at } [x, n]] = x^n$. If the player declines x , he wins iff he wins the sub-game $[x_{n-1}, n-1]$, and $\max(x_{n-1}, \dots, x_1) < x$. We denote the new game state by $[x_{n-1}, n-1, x]$.

Let $V(x, n, x^*) := \Pr[\text{win starting from } [x, n] \cap \max(x, x_n, \dots, x_0) > x^*]$. From principle of dynamic programming, the optimal policy can be yielded from the following system of equations:

$$\begin{aligned} V(x, 0, x^*) &= \mathbf{1}(x > x^*) \\ V(x, n, x^*) &= \begin{cases} \max(x^n, EV(\cdot, n-1, x)) & x \geq x^* \\ EV(\cdot, n-1, x^*) & x < x^* \end{cases} \end{aligned}$$

Our first observation is that a thresholding scheme can be optimal.

Theorem 1. There exists a sequence $T_n \subset \mathbb{R}$ such that $x^n > EV(\cdot, n-1, x) \Leftrightarrow x > T_n$ for all $n \in \mathbb{N}$.

Proof. The LHS of the inequality monotonically increase wrt x , while the RHS monotonically decrease wrt x . Hence the solution set is in the form $(T_n, 1]$. $\square$

We can also say something about T_n :

Theorem 2. The optimal thresholding sequence T_n has the following properties:

1. $T_0 = 0$ and $T_1 = \frac{1}{2}$
2. T_n strictly monotonically increase.
3. $\lim_{n \rightarrow \infty} T_n = 1$

Proof. TBD. □

Now we examine the asymptotic behavior of the optimal policy.

Theorem 3. The optimal policy ensures a $\frac{1}{2}$ winning rate, and the bound is tight for large n . In other words, $V(0, n, 0) \downarrow \frac{1}{2}$.

Proof. TBD. □

Numerical result for T_n is summarized in the following table.

Table 1: Numerical Result for T_n

n	T_n	$EV(\cdot, n, 0)$
0	0.	1.
1	.500	.750
2	.690	.684
3	.776	.655
4	.825	.639
5	.856	.629
6	.878	.622
7	.894	.616
8	.906	.612
9	.916	.609
10	.924	.606

6 Relaxed Problem with Error

Fix $\delta > 0$, and use the following notation for game state: $[\tilde{x}, n]$ means the current item to be decided has observed (inaccurate) value $\tilde{x}$, and we have n items after the current item. If the player examines the current item, the game state evolves into $[x, n-1]$, where we now have the exact value x , and one item (the next) is omitted. If the player otherwise declines the current item, the game state evolves into $[\tilde{x}_n, n-1]$ where $\tilde{x}$ is replaced by $\tilde{x}_n$, the next item, and we have one item less in the queue.

We again obtain the DP equations. $a \wedge b$ denotes $\max(a, b)$.

$$V(x, 0, x^*) = \mathbf{1}(x > x^*) \quad (17)$$

$$V(\tilde{x}, n, x^*) = \max \left(EV(X, n-1, x^* \wedge X_{n-1}), EV(\tilde{X}_{n-1}, n-1, x^* \wedge X) \right) \quad (18)$$

$$V(x, n, x^*) = \begin{cases} \max \left(x^n, E[V(\tilde{X}_{n-1}, n-1, x)] \right) & x \geq x^* \\ E[V(\tilde{X}_{n-1}, n-1, x^*)] & x < x^* \end{cases} \quad (19)$$

The problem with error is again solvable using thresholding:

Theorem 4. The optimal policy for the problem with error can be described completely by thresholding. In other words, for each $\delta \geq 0$, there exists sequences T_n and $R_n \in \mathbb{R}$ such that the LHS in the max operator of (18) is greater than the RHS iff $\tilde{x} > R_n$. Similarly, the LHS is picked in (19) iff $x > T_n$.

Proof. We first prove the existence of T_n . In (19), note that $V(\tilde{X}_{n-1}, n-1, x)$ is monotone decreasing wrt x for each specific realization of $\tilde{X}_{n-1}$. Taking expectation won't affect monotonicity, so the argument in the previous theorem applies.

Now we work on existence of R_n and consider (18). Let $A(x) := E[V(x, n-1, x^* \wedge X_{n-1})]$ and $B(x) := E[V(\tilde{X}_{n-1}, n-1, x^* \wedge x)]$, where the expectation is taken over $(u - \delta, u + \delta) \cap (0, 1)$. Our goal is to prove that $EA > EB$ is a thresholding on u . We will first show that $A(x) > B(x)$ is a thresholding on x , and apply properties of expectation.

Let $x < x^*$. Now $A(x) = E[V(\tilde{X}_{n-2}, n-2, x^* \wedge X_{n-1})]$ and $B(x) = E[V(\tilde{X}_{n-1}, n-1, x^*)]$. $B(x)$ is measure of the event that the best item is selected with value $> x^*$, while $A(x)$ is measure of the same event with additional constraint that X_{n-1} is not the best item. Hence $A(x)$ is a sub-event of $B(x)$, and $A(x) < B(x)$.

Let $x > x^*$ and assume $x < T_n$. Now $A(x) = E[V(\tilde{X}_{n-2}, n-2, x \wedge X_{n-1})]$ and $B(x) = E[V(\tilde{X}_{n-1}, n-1, x)]$. By the same argument, $A(x) < B(x)$.

Now assume $x > T_n$. Take $\omega \in \Omega$. For all $x < X_{n-1}(\omega)$, we have $A_\omega(x) < B(x)$ (why? Need to clarify). For all $x > X_{n-1}(\omega)$, $A_\omega(x) = x^{n-1}$ monotonically increase up to $A(1) = 1$. Hence we have a unique intersection $\alpha \in (T_n, 1)$ where $x > \alpha \Leftrightarrow A_\omega(x) > B(x)$. α does not depend on ω , so it's the unique intersection between $B(x)$ and the "averaged" $A(x)$.

(Another way to proceed) Now assume $x > T_n$. Evaluate the expectation over X_{n-1} , we have

$$A(x) = \int_0^x x^{n-1} dy + \int_x^1 EV(\tilde{X}_{n-2}, n-2, y) dy.$$

As x increases from T_n to 1, the first term monotonically increase from T_n^n to 1, while the second term monotonically decrease and vanishes. $A(x)$ is continuous, so there's at least one intersection. (remains to show uniqueness) $\square$

Again we can say something about the thresholding sequences.

Theorem 5. The optimal thresholding sequences T_n and R_n has the following properties:

1. When $\delta = 0$, $T_n \equiv R_n$ are same as the problem without error.
2. For fixed δ , T_n and R_n both strictly monotonically increase wrt n .
3. For fixed n , T_n and R_n both monotonically decrease wrt δ .
4. $\lim_{n \rightarrow \infty} T_n = 1$
5. (?) $\lim_{n \rightarrow \infty} R_n = 1 - \delta$.
6. As $\delta \rightarrow \infty$, $R_n^{(\delta)} \rightarrow -\infty$ and $T_n^{(\delta)} \rightarrow T_{\lfloor \frac{n}{2} \rfloor}^{(0)}$ for all n .

Proof. TBD. □

Corollary 1. The optimal winning rate $V(0, n)$ satisfies the following:

1. $V_{(0)}$ is identical to the problem without error.
2. As $\delta \rightarrow \infty$, $V_{(\delta)}(0, n) \rightarrow \frac{1}{2} V_{(0)}(0, \lceil \frac{n}{2} \rceil)$ for all n .

We can compute the optimal thresholds by iteratively solving inequalities. Numerical results are summarized below. (Numerical experiments TBD)